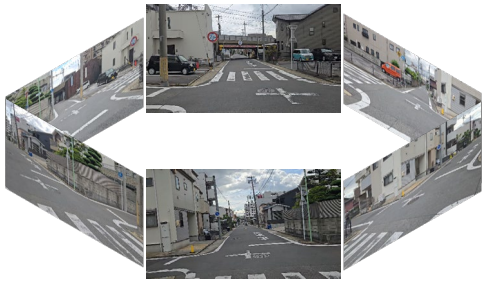
POST-EVENT



Post-event multi-view images

EVENT-TIME



Monocular dashcam video

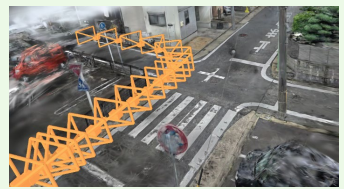
① Static scene and metric alignment

Multi-view SfM → Metric SfM map (+ Road plane) → Static 3DGS



② Dashcam localization

Camera registration



Dashcam trajectory

3. Contact-first human placement

2D body/foot landmarks + temporal contact estimates

↓

Contact-selected foot rays intersect the road

↓

Root refinement: temporal + 2D reprojection

Fixed body pose, orientation, scale, and cameras

OUTPUT



Site 1



Site 2

Free-viewpoint replay